Chapter 2 Teacher Key

Answer Key — Chapter 2 Quiz

- **Artifact:** the matching Chapter 2 page
- **Source alignment:** Reader Ch.2 pp. 7–12.

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
1	Motion is the process of an object changing position	—	Reader p.7	motion = speed	1 pt	—
2	A moving object has energy of motion	—	Reader p.9	energy of motion = speed	1 pt	—
3	moving object = the rolled marble; change = the target marble moved	correct mover + change	Reader p.8	—	2 pts (1 each)	—
4	Energy is the ability to cause change	—	Reader p.1 (Ch.1 spiral, also notes N2)	energy as a substance	1 pt	—
5	BEFORE: target still. AFTER: target moving	any two-frame sketch of that change	Reader p.9 · notes N2	—	2 pts	art style
6	We know because the moving object causes a change. That change is the	any response that uses a change as the evidence	Reader p.9 · notes CHECK · slides 7 and 24–	"I saw the energy"	2 pts	informal wording

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	evidence.		25			
7	The swinging hammer drives the nail into the wood	any useful change the hammer causes	Reader p.10 · notes N3	—	2 pts	a drawing instead of words
8	No. The book is still and nothing changes, so this is not energy of motion	correct non-example	Reader p.9	every object has energy of motion	1 pt	—

Out-of-scope confirmation

- No speed/fast-slow · no ramp height · no mass · no transfer/transformation · no collisions · no kinetic/potential · no engineering.
- Item 6 asks for change-as-evidence only.

Total

12 points (1+1+2+1+2+2+2+1). Scored work is reported as raw score + percentage.

Answer Key — Chapter 2 Practice

- **Artifact:** the matching Chapter 2 page
- **Source alignment:** Reader Ch.2 pp. 7–12; TG L3; AP 3.1.

Part A — What is moving?

Scenario	Object in motion
A basketball falls through the hoop	the basketball
A marble rolls toward another marble	the rolled marble

Scenario	Object in motion
Wind spins a turbine blade	the moving air (wind)
A shopping cart bumps a box	the shopping cart

Part B — What changed?

Scenario	Change caused
The basketball hits the net	the net moves (still → moving)
The marble hits the target marble	the target changes position
Wind spins the blades	the blades rotate (still → spinning)
The cart bumps the box	the box slides forward

Part C — Evidence (which change shows the moving object had energy?)

Observation	Evidence?	Why
"The target marble changed position."	YES	a change happened caused by the mover
"The marble is round."	NO	shape is not a change caused by motion
"The block slid forward after the ball hit it."	YES	still → moving is a change
"I think the ball is heavy."	NO	opinion, not an observed change

Part D — Before / After (sample)

- Before: target block still. After: block moved. Arrows show the direction of the moving object and the target's motion.
- Accept any correct two-frame drawing with a clear change.

Part E — Explain (sample)

"A moving object has energy of motion. When it bumps another object and that object changes position, the change is evidence that the moving object had energy."

- Accept any response that (a) states a moving object has energy of motion, (b) names an observed change, (c) links the change to energy as evidence.
- Do NOT require speed/fast-slow language.

Moving Marbles — Teacher Feedback Key

NOT GRADED — FEEDBACK ONLY. The student page has no points. Do not score this extension.

The page is an optional after-quiz activity. It is not Friday work and it is not Thursday's lesson.

What to look for

Student area	A strong response	Acceptable variation
Predict	The still marble will start to move	Any prediction. A wrong guess is fine.
Trial	1, 2, and 3, same setup	Three rolls, or fewer if time ran out
What moved before?	The rolled marble	"the moving marble"
What changed after?	The target marble started to move	"it slid" or "it changed position"
Evidence	The change is evidence that the moving marble had energy	Any wording that uses the change as the evidence
BEFORE	Target marble still; the other marble rolling toward it	A sketch or a sentence
AFTER	Target marble moving	A sketch or a sentence
Explain	The moving marble had energy of motion. The	Informal wording

Student area	A strong response	Acceptable variation
	change is the evidence.	

Do not

- Do not measure distance or speed in Chapter 2.
- Accept any wording that uses the change as evidence.
- Do not treat a wrong prediction as a mistake if the student names the change.

Chapter 2 Friday Flex — Teacher Key

Not graded. The student page hides the matches.

Card sort

Object	Change	Evidence
Hammer	The nail goes into the wood	I saw the nail move into the wood
Pendulum	The clock gears turn	I saw the gears move
Wind	The blades spin	I saw the blades start to turn

Accept the same matches in a student's own words when the change is the evidence.

Mystery files

Story	Moving object	Change	Evidence
Rolling cart	the cart	the box slides	the box was still, then it moved
Swinging door	the door	the ball rolls	the ball was still, then it moved
Toy car	the car	the blocks fall	the stack was standing, then it fell